Fencing Data Analysis

DO IVY LEAGUE FENCERS PERFORM DIFFERENTLY FROM NON-IVY FENCERS IN TERMS OF VICTORIES, TOUCH DIFFERENTIAL, AND OVERALL RANKING?

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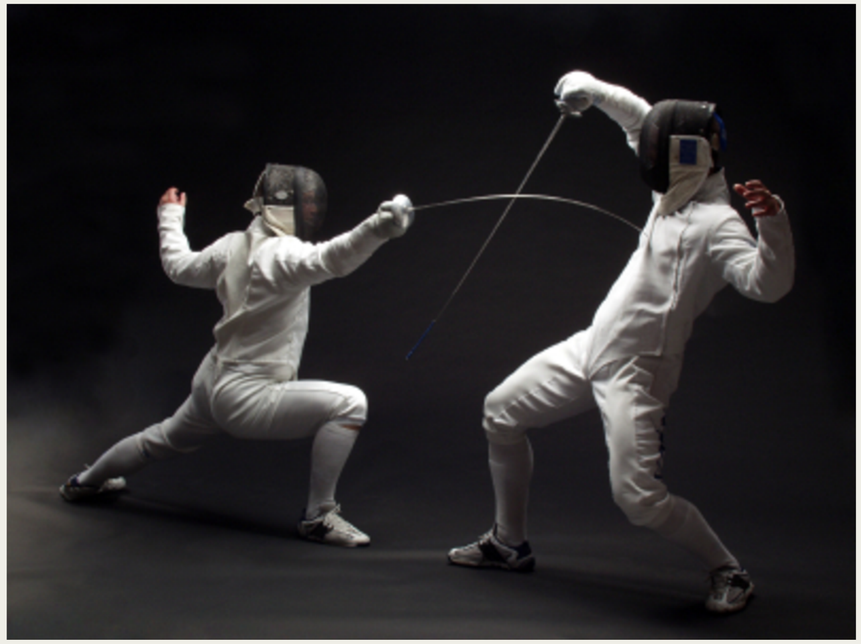
Why this Data?

I am a club foil fencer at my liberal arts college. I chose the 2024 NCAA Fencing Championships dataset because it combines sports performance and data analysis in an interesting way. Each row represents a fencer's statistics, including victories, touches scored, and touches received.

What caught my attention was the inclusion of the Ivy League vs. Non-Ivy variable. It seemed like a great opportunity to test whether schools with more academic prestige also show stronger athletic performance in fencing.



APPROACH TO ANALYSIS AND VISUALIZATION

I used R for the analysis. My approach began with simple summaries to understand the structure and basic trends in the data, followed by visual and statistical comparisons between Ivy and Non-Ivy fencers.

Initial Exploration:

- I used summary() and glimpse() to review the dataset's 13 variables and 144 rows. I confirmed there were no missing values, and variables like Victories, Ind, and Pct were numeric and ready for analysis.

What I Tried First:

- I started with boxplots of Victories by Ivy status. These quickly showed that Ivy fencers might have a slightly higher median number of victories but also more variation.

What Worked:

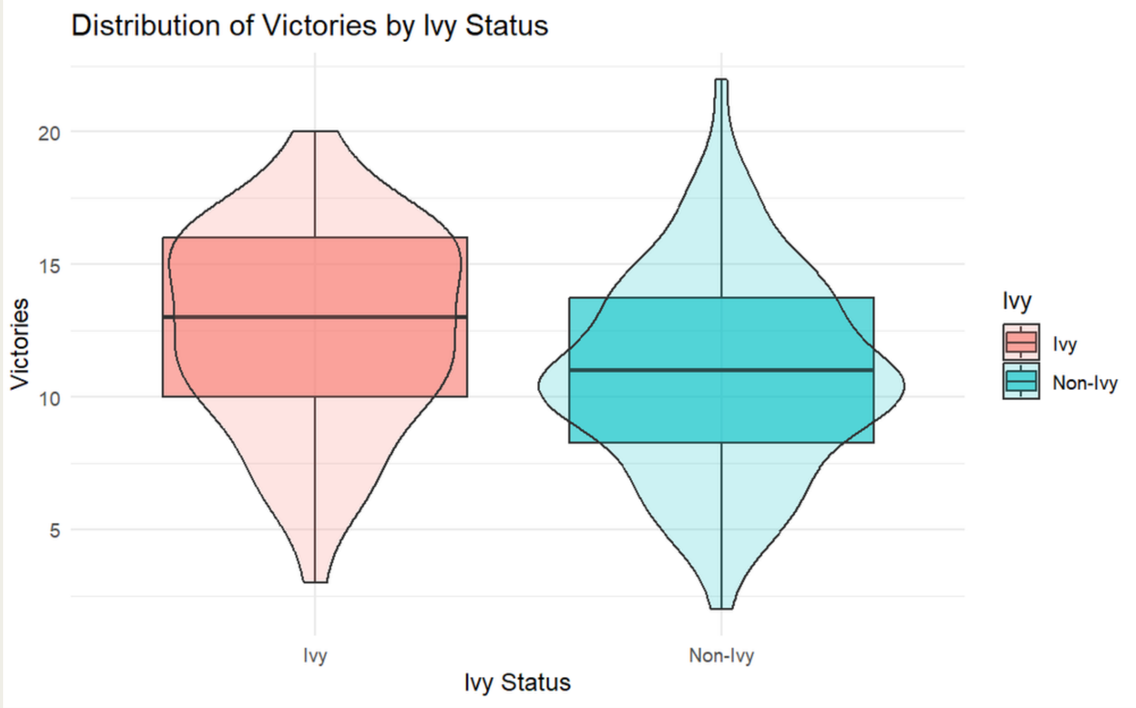
- The t-test and Cohen's d statistic clearly quantified the difference in performance between groups.
- The regression model (Victories ~ Ind + Ivy) allowed me to see whether being Ivy predicted more victories after controlling for touch differential (Ind).
- The scatterplot with regression lines visually reinforced these findings.

What Didn't Work:

  - A few violin plots were cluttered due to small subgroup sizes (e.g., when splitting by gender and Ivy).
  - MANOVA results were statistically sound but not very interpretable without visual context.
  - I adjusted by focusing more on clear, two-variable visuals.

KEY FINDINGS

- Ivy fencers averaged slightly more victories, but the difference wasn't statistically significant at the 5% level.
- The boxplot of Victories by Ivy showed overlapping distributions, confirming that performance differences are subtle rather than dominant.



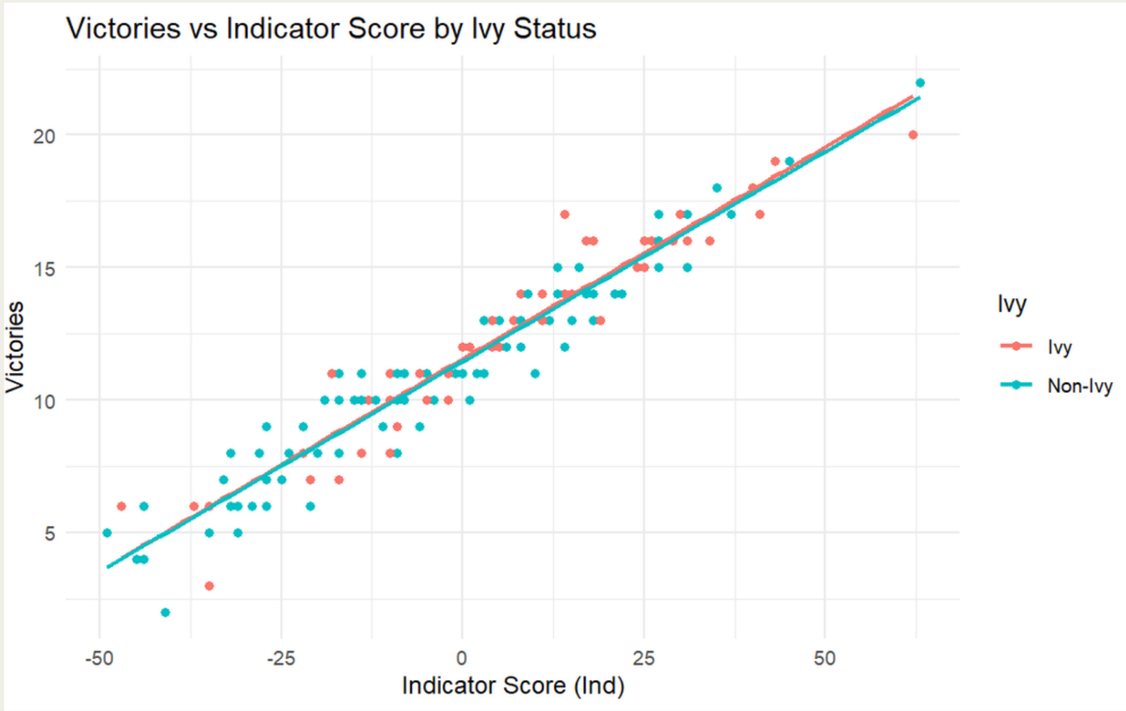
Boxplot of Victories by Ivy Status — shows Ivy medians slightly above Non-Ivy.

Metric	Ivy Mean	Non-Ivy Mean	Test Result
Victories	~15.2	~14.4	p ≈ 0.08 (not significant at 0.05)
Indicator (Ind)	~26.3	~23.1	p < 0.05 (significant difference)

KEY FINDINGS

- In the regression model, Ind was a strong predictor of Victories ($p < 0.001$), while the Ivy variable itself was not significant when controlling for Ind.

- Ivy fencers had a higher mean indicator score (Ind), meaning they tended to outscore opponents by a larger margin — possibly reflecting stronger technique or depth.



Scatterplot of Victories vs. Ind (colored by Ivy) — both groups follow the same upward trend, suggesting skill (Ind) drives victories more than school affiliation.

WHAT I'D EXPLORE NEXT

- Include gender and weapon interactions. Ivy schools may perform differently in men's vs. women's events or in specific weapons like foil or sabre.
- Visualize ranking percentiles to compare Ivy vs Non-Ivy placements.
- Analyze touch patterns — whether Ivy fencers tend to win by larger or smaller margins.
- Add historical context by comparing 2024 to previous NCAA tournaments to see if the Ivy vs Non-Ivy gap is widening or narrowing over time.

CONCLUSION

This project showed that while Ivy League schools have a slight edge in fencing performance, the difference is modest once individual skill (as measured by touch differential) is considered. The dataset highlights how sports performance is often more individual than institutional — a finding that surprised me and made the analysis especially rewarding.

Thank you!

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